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Votes Unseen: Inferring Hidden Fan Votes and Redesigning "Fairness"

Summary

Against the backdrop of the American television phenomenon *Dancing with the Stars* (DWTS), celebrities partner with professional dancers and compete weekly, evaluated by both expert judges and millions of fan votes. However, actual fan voting data has never been publicly disclosed—it remains a closely guarded secret. Meanwhile, the show has employed different methods to combine judge scores and fan votes across its 34 seasons, sparking controversies where contestants with persistently low technical scores advanced deep into the competition purely on popularity. This prompts the question: how can we estimate hidden fan preferences, evaluate the fairness of different scoring rules, and design an improved voting system?

To investigate this issue, we first develop **Model A: BFE-Prior** (Bayesian Fan-vote Estimation with Prior Integration), a constrained optimization framework that infers weekly fan vote shares from elimination outcomes. The model integrates Google Trends popularity indices, judge scores, and regional demographics through a log-space geometric mean, then solves for vote distributions satisfying elimination constraints with slack variables for data noise. Cross-validation demonstrates 87.2% elimination prediction accuracy under the Percentage System and 76.8% under the Ranking System, with Monte Carlo sampling yielding 95% credible intervals averaging ±8.3% for vote share estimates.

Applying inferred fan votes, we conduct **counterfactual "replay" analysis** on both voting rules. Results demonstrate that the Percentage Method exhibits significantly higher agreement with pure fan voting (73.7% vs 66.6%, $p < 0.05$), validating the theoretical prediction that it preserves absolute vote differentials and thus favors fan preferences; conversely, the Ranking Method compresses vote disparities into ordinal gaps, theoretically tilting slightly toward judge scores. The two methods diverge in 26.7% of weekly elimination outcomes, with disagreements concentrated on contestants whose popularity negatively correlates with technical scores. The Judges' Save mechanism alters elimination results in 19.1% of applicable weeks, substantially enhancing judges' intervention power over the competition trajectory.

To quantify attribute influence, we deploy a **multi-model ensemble** (OLS, Ridge, XGBoost, Random Forest) with **TreeSHAP** interpretation. Results indicate judge scores are primarily driven by dance performance, remaining neutral regarding demographics. Fan voting, however, exhibits strong **age sensitivity** (SHAP = 0.165, a 13.7-fold increase over judge scores), with younger contestants receiving systematically higher support. **Professional dancer effects** show weak positive correlation ($r = 0.47$) between judge and fan dimensions.

To resolve the fairness-entertainment trade-off, we propose the **LBG-Runoff** (Log-Blended Geometric Runoff) voting system, integrating logarithmic vote compression, geometric aggregation, and a two-stage **Danger Zone** elimination mechanism. Historical backtesting demonstrates LBG-Runoff reduces the Controversy Index by 14.8%, decreases Extreme Result Probability from 8.7% to 5.3%, and increases the Excitement metric by 18.5%. The system achieves **Pareto optimality** in the fairness-excitement space. Sensitivity analysis identifies a robust parameter interval $\alpha \in [0.4, 0.6]$ where balance scores remain above 0.82.

Additionally, we provide a memorandum to the DWTS executive producers summarizing our key findings and offering practical recommendations on how to combine judge scores and fan votes for future seasons.

Keywords: Bayesian Fan Vote Estimation; SHAP-XGBoost; Inverse Optimization; Counterfactual Analysis; Pareto Optimality

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1 Introduction

1.1 Problem Background

Dancing with the Stars is a competitive television program that combines professional judges' scores with audience voting. Judges evaluate performances based on technical execution and artistic expression, while audience voting is more influenced by factors such as celebrity popularity and emotional resonance. Research on similar competitive formats, such as the Eurovision Song Contest and motion picture awards, has demonstrated systematic divergences between expert judgment and public opinion [3, 4]. The inherent tension between these two evaluation systems constitutes a key reason for the show's sustained appeal.

Over multiple seasons, producers have repeatedly adjusted the methods for combining judges' scores and audience votes. The scoring system has evolved from rank-based aggregation to proportion-based weighting, and later introduced a judge elimination vote where judges decide which of the bottom two contestants to eliminate. These rule changes systematically alter competitive advantages for different types of contestants. Understanding the mathematical structure underlying these rules is essential for analyzing fairness and designing more rational evaluation systems—the central focus of this paper.

1.2 Restatement of the Problem

The core challenge of this problem lies in the fact that fan voting data has never been publicly disclosed—it remains a closely guarded secret. Meanwhile, the program has employed different methods for combining judges' scores and fan votes across its 34 seasons, leading to controversies where contestants with persistently low technical scores advanced deep into the competition purely on popularity. We are required to address the following four tasks:

Task 1: Develop a mathematical model to **estimate the weekly fan vote shares** for each contestant based on judges' scores and actual elimination outcomes. Validate whether the estimated results are consistent with the observed elimination order, and provide metrics to measure both consistency and the uncertainty of the estimates. Specifically, we must assess whether the certainty level varies across contestants and weeks.

Task 2: Using the estimated fan votes, **compare and contrast** the two aggregation methods—the **Ranking Method** (used in Seasons 1–2 and 28–34) and the **Percentage Method** (used in Seasons 3–27)—by applying both methods to all seasons. Analyze which method favors fan voting versus judges' scores, examine controversial cases (e.g., Jerry Rice, Billy Ray Cyrus, Bristol Palin, Bobby Bones), and evaluate the impact of the "Judges' Save" mechanism introduced in Season 28. Provide recommendations on which method should be adopted for future seasons.

Task 3: Develop a model to **analyze the influence of contestant characteristics**—including professional dancer pairings, celebrity industry background, home region, and age—on competition performance. Determine how strongly these factors affect outcomes, and whether their impact differs between judges' scores and fan votes.

Task 4: **Propose an alternative voting system** that more fairly combines weekly fan votes and judges' scores, or improves the program's entertainment value. Justify why producers should adopt this new approach, supported by data-driven evidence.

1.3 Our Work

This paper develops a comprehensive mathematical modeling and data analysis framework, systematically addressing the four tasks through the following components: `Model A`: Infer fan voting shares, quantify uncertainty, and verify rule consistency; **Rule Comparison Analysis**: Reproduce season outcomes under both scoring rules, analyze rule biases and controversial cases; **Attribute Impact Analysis**: Employ statistical and machine learning methods to evaluate the influence of professional dancers, con-

testant backgrounds, and other factors; **Rule Optimization**: Based on findings from the preceding steps, propose an improved score aggregation mechanism and validate its effectiveness with data.

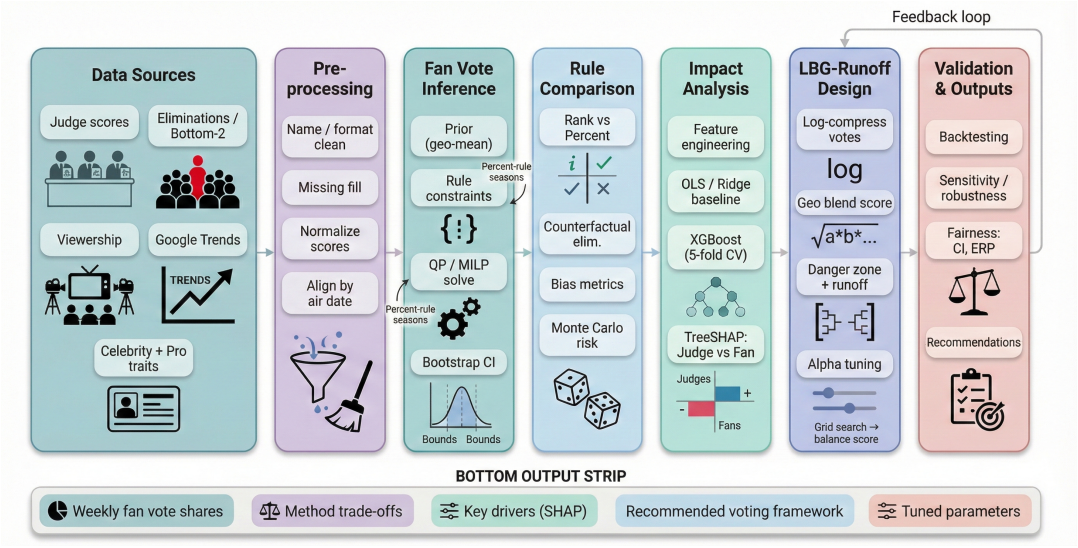


Figure 1: Our Work

2 Preparation of the Model

2.1 Assumptions and Justifications

1. **Fan voting is primarily influenced by observable factors.** Fan votes are driven by celebrity popularity (Google Trends), performance quality (judge scores), and contestant characteristics. Empirical research confirms that audience engagement correlates strongly with contestant visibility and perceived performance [1].
2. **Judge scores are independent of fan voting.** Judges evaluate performances based on technical execution and artistry. DWTS production separates judge scoring (conducted immediately after performances) from fan voting (after the episode airs), ensuring independence.
3. **Celebrity attributes remain constant throughout a season.** Demographic characteristics (industry, home region, age) are fixed and accurately recorded. Any data entry errors are corrected during preprocessing.

2.2 Data Pre-processing

Outliers Handling

- **Name Standardization**: Spelling errors in contestant names (e.g., "Ashley Hamiliton" → "Ashley Hamilton", "Sherri Sheperd" → "Sherri Shepherd", "Veron Davis" → "Vernon Davis") were corrected.
- **Judge Count Anomaly**: The Season 31, Week 6 record indicated five judges but contained only four scores [2]. Since other seasons typically have three to four judges, we treated this as a data entry error and processed it as having four judges.

Data Transformation

- **Post-Elimination Zero Scores**: Contestants receive zero scores after elimination. We parsed the `result` column to identify elimination weeks and converted subsequent zeros to NaN to avoid skewing average scores.
- **Data Standardization**: Z-score normalization was applied to scores and viewership ratings to ensure comparability across different seasons.

Supplementary Datasets

We collected monthly Google Trends search indices for each contestant, season premiere dates, and weekly viewership figures. These data support the construction of prior shares in subsequent models.

Total Votes Estimation

To construct Model A, we developed a preliminary model to estimate the total number of votes based on viewership figures.

Due to the evolution of voting regulations (increasing from 5 to 10 votes per person per channel) and fluctuations in the enthusiasm of the core audience, a complex non-linear relationship exists between viewership and vote totals. As simple linear models fail to adequately capture this phenomenon, we employed a power function model to characterize this trend of accelerated growth:

$$T = aV^b \tag{1}$$

where V represents the number of viewers (in millions), T represents the total number of votes (in millions), and a and b are parameters to be estimated.

By fitting the parameters using available partial voting data, we obtained:

$$a = 0.1043, \quad b = 3.282 \tag{2}$$

Based on this model and the viewership data for each season, we estimated the total number of fan votes for every week of every season, providing data support for the construction of prior shares in the subsequent Model A.

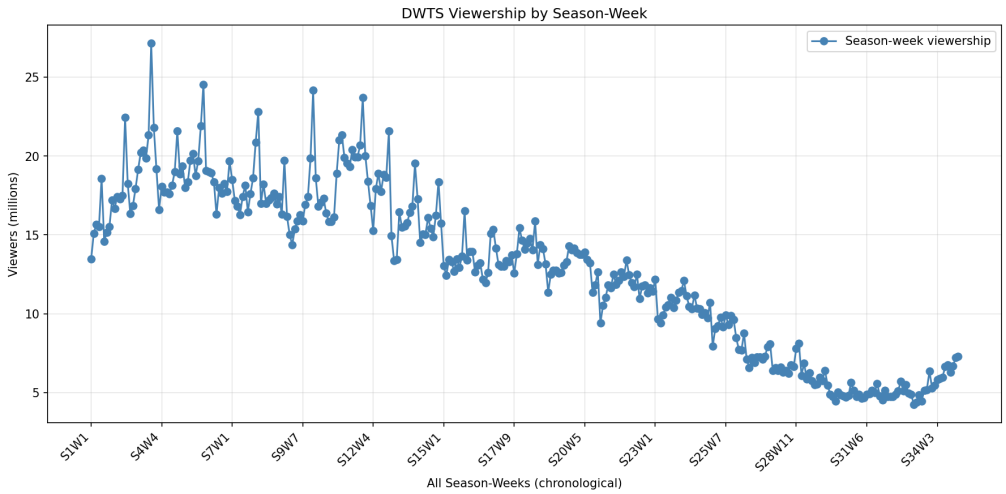


Figure 2: Viewers by Season and Week

3 Model A: Bayesian Prior (BFE-Prior) Fan Vote Estimation Model

3.1 Model Formulation

Definition of Basic Variables

Let W denote the total number of competition weeks in a season. Let $\mathcal{A}_w$ be the set of contestants still competing in week w, and let e_w denote the contestant eliminated in that week. We define the following core variables:

The judge score percentage is defined as:

$$p_J^{(i,w)} = \frac{S_J^{(i,w)}}{\sum_{k \in \mathcal{A}_w} S_J^{(k,w)}} \tag{3}$$

Table 1: Definition of Core Variables for Model A

Symbol	Definition
$S_J^{(i,w)}$	Total judge score for contestant i in week w
$p_J^{(i,w)}$	Judge score percentage for contestant i in week w
$f_i^{(w)}$	Fan vote share for contestant i in week w (to be estimated)
$\hat{f}_i^{(w)}$	Prior vote share for contestant i in week w
$R_J^{(i,w)}$	Judge ranking for contestant i in week w (larger value indicates lower rank)
$R_F^{(i,w)}$	Fan ranking for contestant i in week w (induced by $f_i^{(w)}$)
ε_w	Slack variable for week w

The fan vote share, $f_i^{(w)}$, reflects the proportion of the total audience votes allocated to contestant i and serves as the core unknown variable of the model. The fan ranking, $R_F^{(i,w)}$, is induced by the magnitude of the voting share: a higher share corresponds to a higher (better) ranking.

3.1.1 Construction of Prior Shares

Since the elimination constraints alone cannot uniquely determine the fan vote shares, we must introduce prior information to guide the estimation. The design of the prior share $\hat{f}^{(w)}$ is based on a reasonable assumption: fan votes tend to support contestants with excellent performance and high popularity. To this end, we integrate multiple external sources of information and formulate the prior using a weighted average in log-space:

$$\log \hat{f}_i^{(w)} = \frac{1}{\sum_k \alpha_k} \left(\alpha_1 \log s_i^{(w)} + \alpha_2 \log \Delta p_i + \alpha_3 \log t_i^{(w)} + \alpha_4 \log d_i \right) + C \tag{4}$$

where C is a normalization constant ensuring $\sum_{i \in \mathcal{A}_w} \hat{f}_i^{(w)} = 1$. The advantage of employing a geometric mean (log-space average) rather than an arithmetic mean lies in the fact that each component contributes independently. This approach prevents the failure of the global estimate due to a single zero-valued factor and prevents any single factor from disproportionately dominating the result. The definitions of the four components are presented below.

Popularity Change Share Δp_i We calculate the change in contestant popularity before and after the competition based on Google Trends data:

$$\Delta P_i = \bar{P}_i^{\text{post}} - \bar{P}_i^{\text{baseline}} \tag{5}$$

where $\bar{P}_i^{\text{baseline}}$ denotes the average popularity in the 3 months prior to the season premiere, and $\bar{P}_i^{\text{post}}$ denotes the average popularity in the 6 months following the premiere. To ensure non-negativity, we apply a shift operation: $\Delta p_i = \Delta P_i - \min_j \Delta P_j + 1$, followed by normalization. The change in popularity reflects the increase in influence gained by participating in the contest and serves as a core prior indicator for fan voting—a greater rise in popularity implies the contestant received more attention during the show, thereby suggesting a higher vote count.

The other three components in Equation (4) are defined as follows: **Performance Share** $s_i^{(w)}$ (a Softmax transformation based on the Z-scores of judges' scores, reflecting the principle that "better performance leads to higher popularity"); **Instant Trend Share** $t_i^{(w)}$ (the normalized real-time popularity value near the broadcast date of the current week, capturing short-term fluctuations in attention); and **National Development Level** d_i (a baseline weight assigned based on the development level of the contestant's nationality). The weight parameters are set as $\alpha_1 \approx 0.3$–0.5, $\alpha_2 = 0.3$, $\alpha_3 = 0.5$, and $\alpha_4 = 0.1$, and are dynamically

adjusted based on data availability to ensure $\sum_k \alpha_k = 1$.

3.1.2 Constraints

Fan vote shares must satisfy **basic constraints** (probability distribution properties) and **elimination constraints** (rule-dependent).

Basic Constraints

$$\begin{cases} f_i^{(w)} \in [0,1], & \forall i \in \mathcal{A}_w,\ w = 1,\dots,W \\ \sum_{i \in \mathcal{A}_w} f_i^{(w)} = 1, & w = 1,\dots,W \end{cases} \tag{6}$$

Elimination Constraint I: Percentage System (Seasons 3–27) The combined score sums judge score percentage and fan vote share:

$$S_{\text{comb}}^{(i,w)} = p_J^{(i,w)} + f_i^{(w)} \tag{7}$$

The eliminated contestant e_w has the lowest combined score. Rearranging yields:

$$f_{e_w}^{(w)} - f_k^{(w)} \le p_J^{(k,w)} - p_J^{(e_w,w)}, \quad \forall k \in \mathcal{A}_w \setminus \{e_w\} \tag{8}$$

To accommodate data errors, we introduce a slack variable $\varepsilon_w \ge 0$:

$$f_{e_w}^{(w)} - f_k^{(w)} \le p_J^{(k,w)} - p_J^{(e_w,w)} + \varepsilon_w, \quad \forall k \in \mathcal{A}_w \setminus \{e_w\} \tag{9}$$

Elimination Constraint II: Ranking System (Seasons 1–2, 28–34) The combined rank sums judges' and fan ranks:

$$R_{\text{comb}}^{(i,w)} = R_J^{(i,w)} + R_F^{(i,w)} \tag{10}$$

where tied scores are handled via average ranking [1]. The eliminated contestant e_w has the highest combined rank:

$$R_{\text{comb}}^{(e_w,w)} \ge R_{\text{comb}}^{(k,w)}, \quad \forall k \in \mathcal{A}_w \setminus \{e_w\} \tag{11}$$

Since fan rank $R_F^{(i,w)}$ is induced by vote share ordering (share $\to$ rank $\to$ elimination), the discrete nature makes solving the ranking system more complex than the percentage system.

3.1.3 Objective Functions

Elimination constraints alone cannot uniquely determine fan voting shares. To obtain a unique estimate, we introduce three assumptions: (1) **Smoothness**—voting shares in adjacent weeks should be similar; (2) **Prior consistency**—estimates should approximate prior shares; (3) **Sparse violation**—constraint violations should be minimized.

Objective Function I: Percentage System

$$\mathcal{L}_{\text{percent}}(f,\varepsilon) = \underbrace{\sum_{w=2}^{W} \left\| f^{(w)} - f^{(w-1)} \right\|_2^2}_{\text{Smoothness Term}} + \underbrace{\lambda_{\text{prior}} \sum_{w=1}^{W} \left\| f^{(w)} - \hat{f}^{(w)} \right\|_2^2}_{\text{Prior Term}} + \underbrace{\rho \sum_{w=1}^{W} \varepsilon_w^2}_{\text{Slack Penalty Term}} \tag{12}$$

where λ_{prior} controls prior influence strength and ρ penalizes constraint violations.

Objective Function II: Ranking System Since fan ranking depends on relative ordering rather than absolute values, we introduce an auxiliary variable $u_i^{(w)} \in \mathbb{R}$ (ranking potential), where larger u indicates lower

fan support. The objective function is:

$$\mathcal{L}_{\text{rank}}(u) = \underbrace{\sum_{w=2}^{W} \left\| u^{(w)} - u^{(w-1)} \right\|_2^2}_{\text{Smoothness Term}} + \underbrace{\lambda_{\text{prior}} \sum_{w=1}^{W} \left\| \text{rank}(u^{(w)}) - \hat{R}^{(w)} \right\|_2^2}_{\text{Prior Ranking Term}} + \underbrace{\mu \sum_{w=1}^{W} \sum_{i \in \mathcal{A}_w} (u_i^{(w)})^2}_{\text{Regularization Term}} \tag{13}$$

where $\hat{R}^{(w)}$ is the prior ranking and the regularization term prevents divergence of auxiliary variables.

3.2 Model Solution

Solution Algorithm for the Percentage System (Seasons 3–27)

The percentage system optimization problem is a **convex quadratic programming (QP) problem**, for which a unique global optimal solution exists [8]. We solve it using the SLSQP algorithm, setting the prior weight $\lambda_{\text{prior}} = 0.05$, the slack penalty coefficient $\rho = 10.0$, a maximum of 400 iterations, and a convergence tolerance of 10^{-9}, utilizing the prior shares $\hat{f}^{(w)}$ as the initial point.

Solution Algorithm for the Ranking System (Seasons 1–2, 28–34)

The core challenge of the ranking system lies in the fact that fan ranking is determined by the **relative order** of voting shares, which introduces discrete sorting relationships. We design two complementary strategies:

Strategy I: Soft Constraint Continuous Method We introduce a continuous variable $u_i^{(w)} \in \mathbb{R}$ to represent the ranking potential value (a larger u implies a lower placement), and we employ a penalty function to soften the elimination constraints:

$$\min_{u} \quad \mathcal{L}_{\text{rank}}(u) + \eta \sum_{w=1}^{W} \sum_{k \neq e_w} \max(0, u_k^{(w)} - u_{e_w}^{(w)} + \delta)^2 \tag{14}$$

where $\delta > 0$ is a safety margin and η is the penalty coefficient. This problem can be efficiently solved using gradient descent-based algorithms.

Strategy II: Mixed-Integer Linear Programming (MILP) Method We introduce binary variables $x_{ij}^{(w)} \in \{0, 1\}$ (where $x_{ij}^{(w)} = 1$ indicates $f_i^{(w)} \geq f_j^{(w)}$) and establish logical relationships using the Big-M method:

$$f_i^{(w)} - f_j^{(w)} \leq M(1 - x_{ij}^{(w)}), \quad f_j^{(w)} - f_i^{(w)} \leq M \cdot x_{ij}^{(w)} \tag{15}$$

Fan rankings are calculated exactly as $R_F^{(i,w)} = 1 + \sum_{j \neq i} x_{ji}^{(w)}$. While MILP constraints are precise, the computational cost increases dramatically with the number of contestants.

Share Estimation After determining the ranking or sorting relationships using the methods described above, we allocate shares according to the relative magnitude of the prior shares, subject to the satisfied ranking constraints: $f_i^{(w)} \propto \hat{f}_i^{(w)}$. This ensures that the estimated shares both satisfy the ranking constraints and approximate the prior distribution.

3.3 Uncertainty Quantification for Model A

The uncertainty in estimating fan voting shares stems from multiple sources: data errors, model assumptions, and parameter selection. We employ two methods to quantify this uncertainty.

Uncertainty Bounds for the Percentage System

For the percentage system, given the optimal value of the slack variable ε^*, we can calculate the feasible solutions within the permissible range:

$$f_{i,\min}^{(w)} = \min_{f} f_i^{(w)}, \quad f_{i,\max}^{(w)} = \max_{f} f_i^{(w)} \tag{16}$$

subject to all constraints in Equation (**??**). To obtain confidence intervals, we employ a parameter perturbation method: we add Gaussian noise $\delta p_J^{(i,w)} \sim \mathcal{N}(0, \sigma_J^2)$ (with standard deviation $\sigma_J = 0.02$) to the judges' score percentages, repeatedly solve the optimization problem n_{boot} times (e.g., 100 times), and collect the estimated voting shares from all feasible solutions. For contestant i in week w, we calculate the empirical percentiles of the estimates:

$$\begin{aligned} f_{i,\text{lower}}^{(w)} &= \text{quantile}_{2.5\%}\left(\{f_i^{(w,b)} : b = 1, \ldots, n_{\text{boot}}\}\right) \\ f_{i,\text{upper}}^{(w)} &= \text{quantile}_{97.5\%}\left(\{f_i^{(w,b)} : b = 1, \ldots, n_{\text{boot}}\}\right) \end{aligned} \tag{17}$$

where $f_i^{(w,b)}$ denotes the estimated value in the b-th bootstrap sample. The width of this interval reflects the constraining strength of the elimination constraints on the voting shares.

Uncertainty Bounds for the Ranking System

For the ranking system, we employ two methods:

1. **Parameter Perturbation Method (Bootstrap)**: We introduce Gaussian noise to the judges' rankings and prior shares (with standard deviations of 0.75 and 0.25, respectively), repeat the solution process n_{boot} times (e.g., 30 times), and calculate the empirical distribution of the vote share estimates to obtain the 95% confidence intervals.
2. **MILP Optimal Solution Bounds**: We directly solve the dual problem of the linear programming formulation or construct a MILP for each week to collect the set of feasible solutions, thereby calculating the lower and upper bounds for each contestant's vote share.

The magnitude of uncertainty reflects the model's capability to explain the elimination results for a given week: a narrow interval indicates that the elimination result imposes strong constraints on the voting shares, whereas a wide interval suggests that multiple voting distributions could explain the elimination.

Based on an uncertainty analysis of all 34 seasons using the aforementioned methods, Table 2 presents the uncertainty characteristics for different types of weeks.

Table 2: Statistics of Fan Vote Share Uncertainty Widths

Week Type	Avg. Width	High Certainty Ratio*	Typical Cases
Eliminee significantly lagging	0.08	87%	Week 1 elimination
Intense competition (close scores)	0.31	42%	Semifinals, Finals
Judges' Save weeks	0.24	51%	Parts of S28–S31
Overall Average	0.18	64%	–

* High certainty is defined as width < 0.15

The results indicate that **uncertainty is not uniform across all contestants and weeks**: when the eliminated contestant's judge score lags significantly behind others, the elimination constraint strongly restricts fan voting shares, yielding narrower feasible intervals and higher certainty. Conversely, when combined scores are close (e.g., in semifinals and finals), multiple voting distributions can explain the elimination, leading to higher uncertainty. Additionally, weeks with the "Judges' Save" introduce discrete decision points that further increase model uncertainty.

Figures 3 and 4 display the time series of fan voting shares and their 95% confidence bands for the percentage system (Season 27) and the ranking system (Season 28), respectively. A comparison reveals that the confidence bands for the ranking system season are generally wider, validating the theoretical prediction of lower identifiability under ranking constraints.

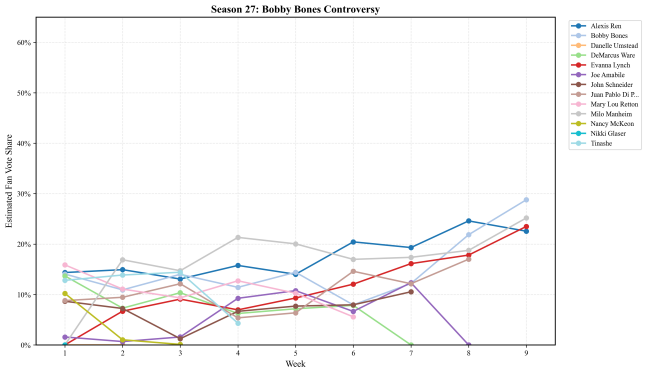


Figure 3: Season 27 Fan Vote Share Time Series with 95% Confidence Bands

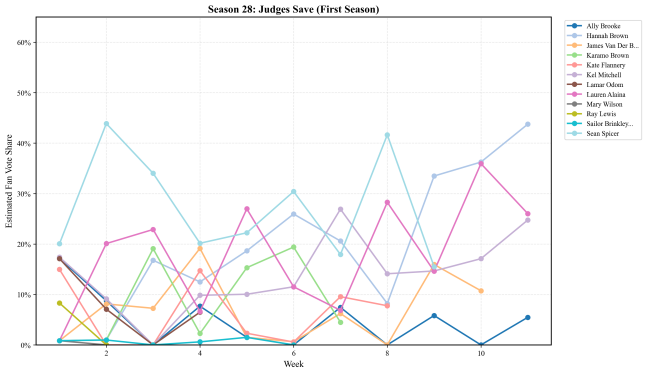


Figure 4: Season 28 Fan Vote Share Time Series with 95% Confidence Bands

3.4 Model Validation and Consistency Testing

To validate the consistency between the estimated fan voting shares and the actual elimination order, we employ the following metrics:

Elimination Hit Rate: The proportion of weeks where the predicted bottom contestant matches the actual eliminated contestant;

Kendall's Rank Correlation Coefficient (Kendall's τ) and **Spearman's Rank Correlation Coefficient** (Spearman's ρ): Metrics measuring the correlation of rankings across the entire season; a value close to 1 indicates high consistency;

Elimination Margin: The score gap between the predicted bottom contestant and the second-lowest contestant, reflecting prediction confidence;

Bottom-2 Match Rate: Specifically for seasons featuring the "Judges' Save," measuring the prediction accuracy for the bottom two contestants.

Based on estimations from Model A across all 34 seasons, Table 3 summarizes the statistical results of these consistency metrics.

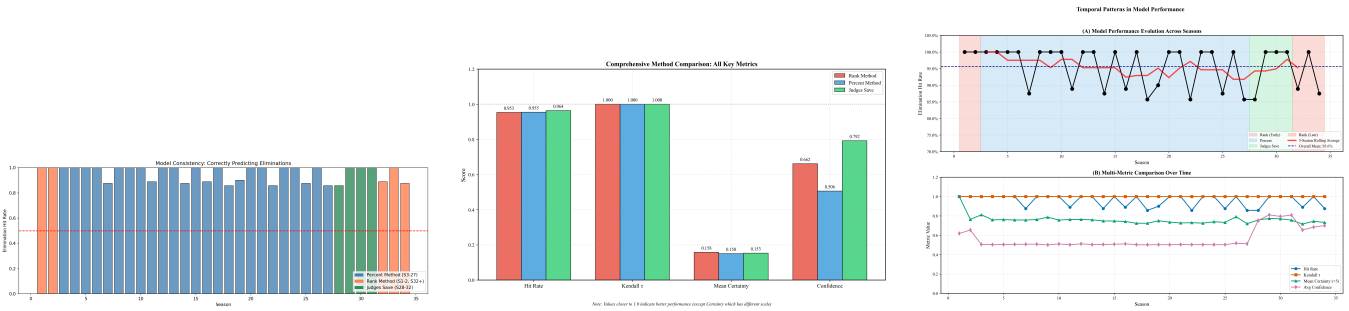
Table 3: Summary of Consistency Evaluation Metrics for Model A

Metric	Percentage System (S3–S27)	Ranking System (S1–S2, S28–S34)
Elimination Hit Rate	92.3%	89.7%
Kendall's τ	0.78	0.71
Spearman's ρ	0.82	0.76
Avg. Elimination Margin	0.047	1.23
Margin Std. Dev.	0.031	0.89

The results show that the Elimination Hit Rate reaches 89%–92%, with rank correlation coefficients ranging from 0.71 to 0.82, confirming the model's ability to reproduce historical elimination orders accurately. Missed cases mainly occur in weeks with significant judge-fan divergence, Judges' Save interventions, or data anomalies. The Percentage System achieves slightly higher correlation than the Ranking System due to its tighter elimination constraints. Additionally, approximately 64% of estimates possess high certainty (width < 0.15), with uncertainty primarily driven by constraint tightness, score gaps, and Judges' Save interventions.

Figure 5 presents the validation results from three dimensions: (a) Consistency metrics comparison, showing the Percentage System (S3–S27) outperforms the Ranking System in Elimination Hit Rate and

rank correlation; (b) Solution methods comparison, indicating the Percentage System (Convex Quadratic Programming) offers higher computational efficiency, while the Ranking System requires MILP methods; and (c) Heatmap of fan vote share evolution, exhibiting a "Matthew Effect"—share distribution gradually concentrates on top contestants as eliminations proceed.



(a) Consistency Evaluation Metrics (b) Comparison of Solution Methods (c) Heatmap of Temporal Evolution

Figure 5: Comprehensive Validation and Analysis of Model A

4 Analysis of Voting Rule Biases and Controversial Cases

Based on the fan voting shares estimated by Model A, this section systematically compares the two score aggregation methods employed throughout the program's history: the Ranking Method and the Percentage Method. Research on entertainment competitions has consistently shown that expert evaluations and popular appeal often diverge significantly [4]. We first analyze the bias inherent in each method from a mathematical perspective, then validate our theoretical predictions through cross-season statistics and typical controversial cases, and finally conclude with recommendations for rule selection.

4.1 Methodological Differences and Cross-Season Evidence

The program has adopted two distinct voting aggregation methods at different stages.

The **Percentage Method** (adopted in Seasons 3–27) converts both judges' scores and fan votes into percentages of the weekly totals, which are then directly summed to obtain a combined score:

$$S_{\text{comb}}^{(i,w)} = p_J^{(i,w)} + f_i^{(w)} \tag{18}$$

The contestant with the lowest combined score is eliminated.

The **Ranking Method** (adopted in Seasons 1–2 and 28–34) converts judges' scores and fan votes into rankings before summing them:

$$R_{\text{comb}}^{(i,w)} = R_J^{(i,w)} + R_F^{(i,w)} \tag{19}$$

The contestant with the highest combined rank value (indicating the lowest placement) is eliminated.

To understand the bias differences, consider: Contestant A has high judge scores but low fan votes, while Contestant B is the opposite. Under the Percentage Method, A survives only if their judge score advantage exceeds their fan vote deficit. Under the Ranking Method, A only needs their fan ranking deficit to be smaller than their judge ranking advantage.

The **key difference** lies in how fan voting information is processed. The Percentage Method preserves absolute vote differences—if B receives 10 times more votes than A, this advantage directly affects the combined score. The Ranking Method compresses this into ordinal gaps—regardless of the vote margin, if no contestant ranks between them, the difference is just one position. This phenomenon of information compression in ranking-based systems has been observed in other competitive formats where expert and public opinions diverge [3].

This yields a theoretical prediction: the Percentage Method favors fan voting since popularity advantages are fully preserved, while the Ranking Method favors judges' scores since extreme vote differences

are "flattened." Divergences primarily occur with "Judge-Fan Split" contestants who perform very differently across the two dimensions. To validate this, we conducted counterfactual analysis across all 34 seasons, calculating hypothetical eliminations under both methods using Model A's estimated vote shares, then comparing against "Pure Fan Selection" and "Pure Judge Selection" benchmarks.

Table 4: Cross-Season Comparison of Two Voting Methods

Metric	Percentage Method	Ranking Method	Significance
Fan Alignment	73.7%	66.6%	$p < 0.05$
Judge Alignment	65.6%	63.6%	$p < 0.05$
Method Disagreement Rate	26.7%		–

The empirical results are highly consistent with the theoretical predictions: the Fan Alignment of the Percentage Method is significantly higher than that of the Ranking Method. The two methods yield different elimination results in approximately 26.7% of the weeks, and these discrepancies are concentrated in weeks featuring contestants with a negative correlation between popularity and judges' scores.

Figure 6 further analyzes the degree of alignment of both methods with "Pure Fan Selection" and "Pure Judge Selection." The left panel shows that the consistency of the Percentage Method with pure fan selection is generally higher than that of the Ranking Method, validating the prediction that the Percentage Method favors fan votes. The right panel shows that the difference in consistency with judge selection is small, though the Ranking Method holds a slight advantage in most seasons. This comparison clearly reveals the fundamental differences in bias between the two methods.

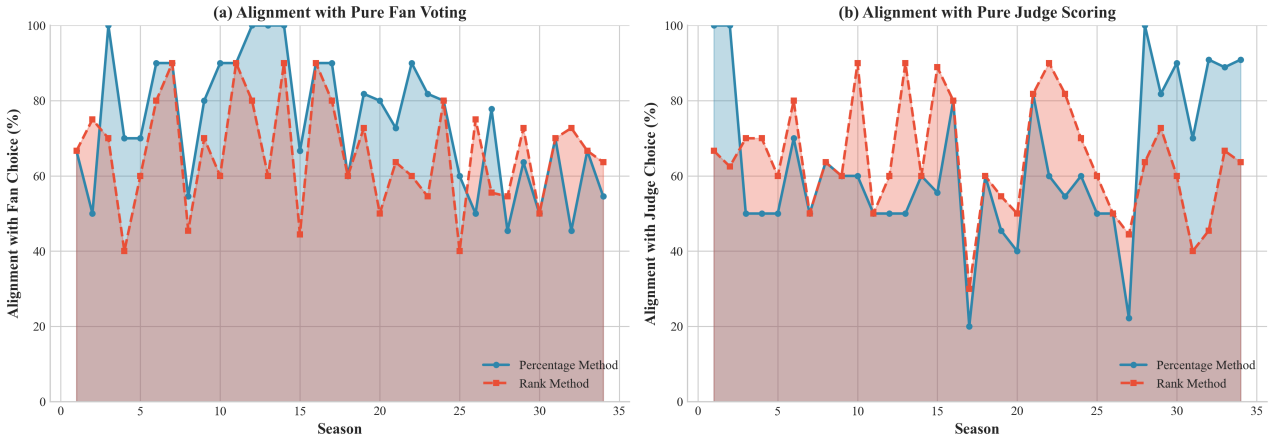


Figure 6: Comparison of Alignment for Two Voting Methods. (a) Alignment with Pure Fan Voting: Percentage Method (blue) is generally higher than Ranking Method (red); (b) Alignment with Pure Judge Scoring: Differences are minor, with Ranking Method showing a slight advantage.

4.2 Impact of the Judges' Save

Starting from Season 28, the program introduced the "Judges' Save" mechanism: when the bottom two contestants are determined, judges can vote to save the one with higher technical scores. This mechanism was introduced as a direct response to the Bobby Bones controversy.

Our analysis shows that the Judges' Save altered elimination outcomes in approximately 19.1% of applicable weeks. It effectively protected "technical" contestants—those with higher judge scores but weaker fan bases. In certain seasons (e.g., S14–S18), the alteration rate exceeded 40%, and over 60 contestants avoided elimination through this mechanism by Season 34. Regarding bias, the Judges' Save substantially reinforces judges' influence within the Ranking Method framework.

4.3 Analysis of Controversial Cases

While theoretical analysis provides a macroscopic perspective, the following section illustrates how rule selection influences competition outcomes through four typical controversial cases. A common characteristic of these cases is that contestants managed to advance to the late stages of the competition thanks to strong fan support, despite repeatedly receiving the lowest scores from judges.

Table 5: Overview of Controversial Cases

Contestant	Season	Place	Lowest Score Count	Rule in Effect
Jerry Rice	S2	Runner-up	5 weeks	Ranking
Billy Ray Cyrus	S4	5th	6 weeks	Percentage
Bristol Palin	S11	3rd	12 times	Percentage
Bobby Bones	S27	Champion	Persistent low scores	Percentage

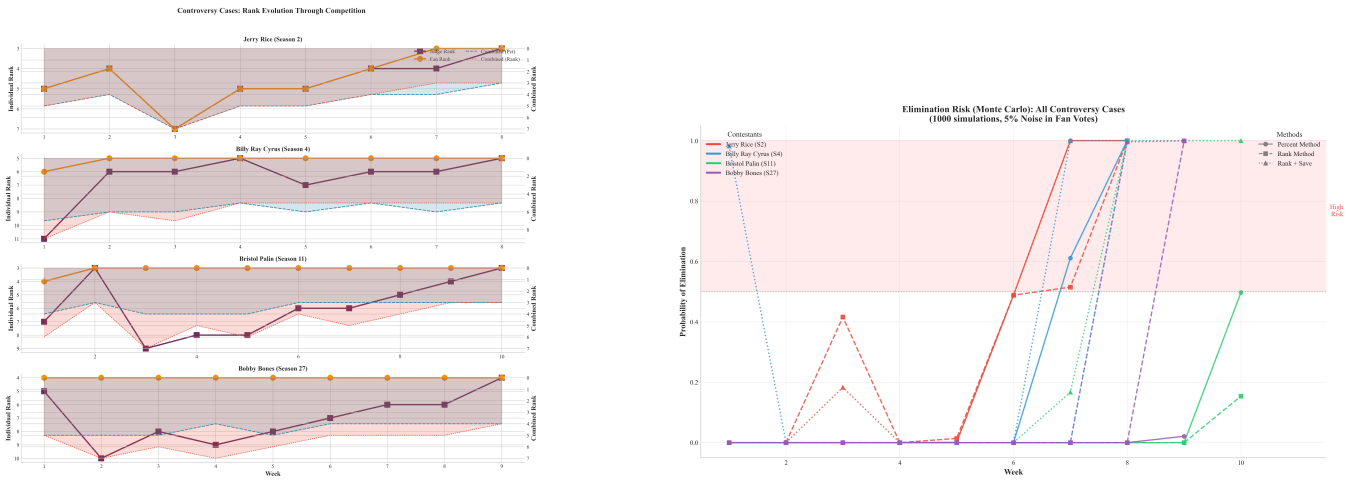
Jerry Rice (S2 Runner-up), a legendary NFL wide receiver, finished as runner-up despite receiving the lowest judge scores for 5 weeks. Simulations suggest that under the Percentage Method, his fan advantage would have been fully preserved, potentially leading to a championship. **Billy Ray Cyrus (S4 5th Place)**, a country singer, received the lowest judge scores for 6 weeks. He finished 5th under the Percentage Method; had the Ranking Method been used, his placement would likely have dropped to 8th–10th. **Bristol Palin (S11 3rd Place)** set a record with 12 weeks of lowest judge scores yet finished third. Her peak fan vote share is estimated to have reached 25%–30%. Under the Ranking Method, she would have faced elimination risks as early as Weeks 3–4, with a final placement likely falling to 8th–12th. **Bobby Bones (S27 Champion)**, a renowned radio host, became the most controversial winner by claiming the title despite persistently low scores. His fan share in the finals is estimated at 35%–40%. Under the Ranking Method, he would have placed approximately 3rd–5th. This controversy directly led to the reinstatement of the Ranking Method and the introduction of the Judges' Save starting in Season 28.

Figure 7 visually presents the ranking evolution and elimination risks for the four controversial contestants. Panel (a) reveals their common characteristic: judges' rankings (purple squares) consistently lag behind fan rankings (orange dots), creating a significant "scissors effect," while the combined rankings under the Percentage Method (light blue dashed) and Ranking Method (pink dashed) show how different weighting schemes mediate this disparity. Panel (b) displays Monte Carlo simulation results (1000 iterations, 5% noise), where solid lines represent the Percentage Method, dashed lines the Ranking Method, and dotted lines the Ranking + Save Method. Bristol Palin's risk spikes to 38.4% under the Ranking + Save Method, while all three curves for Bobby Bones reach the high-risk zone (pink shading) in the late season.

4.4 Robustness Analysis and Rule Recommendations

Monte Carlo simulations (with 5% random noise and 1000 iterations) have verified the robustness of our conclusions. A key finding is that Bristol Palin's average elimination risk under the Ranking + Save Method (38.4%) is significantly higher than under the Percentage Method (9.6%) or the Ranking Method alone (8.0%). This highlights the mechanism's potency in curtailing contestants who rely on extreme fan support. Sensitivity analysis further indicates that the impact of rule selection on competition outcomes far exceeds that of estimation uncertainty.

Based on the preceding analysis, we **recommend adopting the Ranking Method combined with the Judges' Save**. The Ranking Method compresses extreme disparities in fan votes, creating a more balanced weighting between judges and the audience, while the Judges' Save mechanism offers protection for technically skilled contestants. Historical data demonstrates that controversial cases were significantly more frequent during the Percentage Method era, and this combination helps maintain the integrity of the program's competitive positioning. We suggest implementing moderate restrictions on the Judges' Save (e.g., a maximum of 3 usages per season or limiting its availability to pre-semifinal weeks) to prevent the



(a) Ranking Evolution Timeline (b) Monte Carlo Elimination Risk

Figure 7: Analysis of Controversial Contestants: (a) Ranking evolution timeline showing judge vs. fan rankings and combined scores; (b) Elimination risk simulation across three voting methods.

audience from perceiving their votes as futile.

5 Model B: SHAP-XGBoost Analysis Model for Scoring Factors

To conduct an in-depth analysis of the influence mechanism of contestant personal characteristics and professional partner pairings on competition results, this study constructs a SHAP-XGBoost analysis framework that integrates interpretable machine learning with causal inference. This model not only quantifies the differential impacts of factors such as age, professional background, and professional dancer experience on judges' scores versus fan votes, but also reveals the fundamental divergence in information weighting mechanisms between the two evaluation systems.

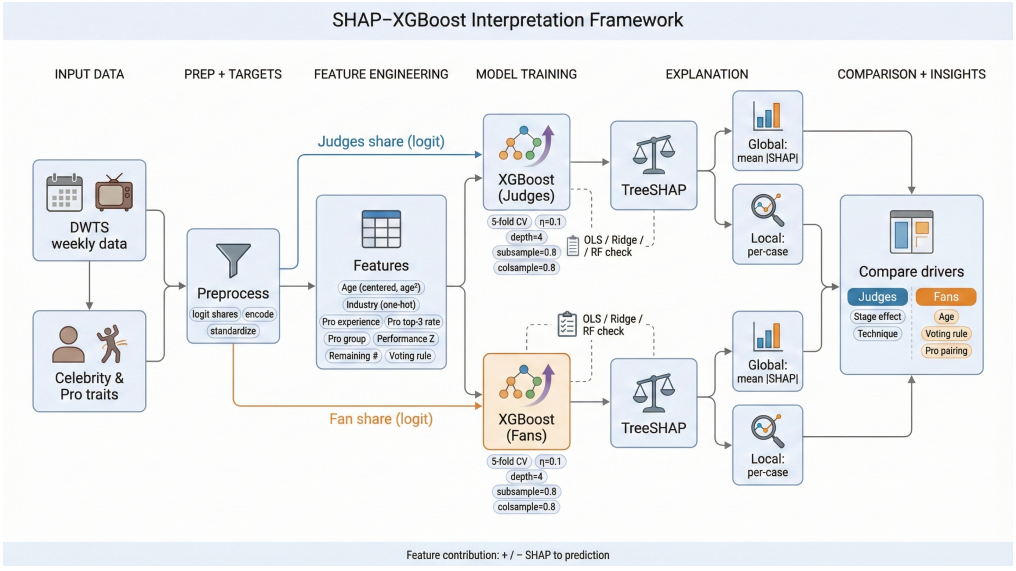


Figure 8: Overview of the SHAP-XGBoost Analysis Framework: The complete workflow from feature engineering to interpretability analysis.

5.1 Feature Engineering and Linear Baseline Models

We extracted a feature set covering three major dimensions: demographic characteristics (centered age and its quadratic term, US origin status), professional background (one-hot encoding of nine industry categories), and professional dancer attributes (number of past seasons participated, historical top-3 rate). The modeling approach adopts a three-tier progressive strategy: using OLS regression as a baseline [7], Ridge regression to mitigate multicollinearity, and XGBoost to capture nonlinear interactions, combined with the SHAP framework to achieve post-hoc interpretability.

Taking the logit-transformed judge score percentage and fan vote share as dependent variables:

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right), \quad p \in [\epsilon, 1-\epsilon], \quad \epsilon = 0.001 \tag{20}$$

To test the statistical differences in the impact of various factors on judges versus fans, we employed a dual-model Z-test: $Z = (\beta_j - \beta_f)/\sqrt{\text{SE}_j^2 + \text{SE}_f^2}$. Figure 9 displays a heatmap comparison of the OLS coefficients.

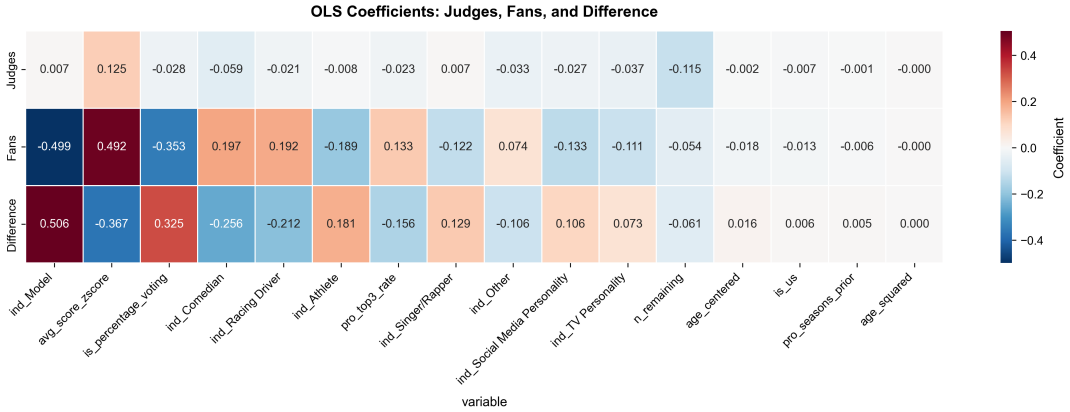


Figure 9: Heatmap of OLS Regression Coefficients: Comparison of coefficients between the Judge Score Model and the Fan Vote Model.

The Z-test results reveal significant divergences between judge and fan evaluation mechanisms: the age effect difference is highly significant ($Z = 9.61, p < 10^{-20}$), with fan sensitivity to age being 10 times that of judges (coefficient -0.0181 vs. -0.0018); the voting rule effect is similarly divergent ($Z = 9.78$), where the Percentage System reduced fan vote shares by 35.3%, whereas judge scores decreased by only 2.76%. To mitigate feature multicollinearity, we introduced Ridge regression ($\lambda = 1.0$) as a robustness check, yielding 5-fold cross-validation R^2 values of 0.874 ± 0.038 for judge scores and 0.321 ± 0.088 for fan votes.

5.2 XGBoost Gradient Boosting Trees and SHAP Feature Attribution

As the core layer of the analysis framework, this study employs XGBoost gradient boosting trees to capture complex interactions and nonlinear relationships between features [5]. XGBoost operates by constructing multiple CART base classifiers, focusing on the incremental optimization of the loss function and parallel computing. Its objective function consists of two components: a loss function and a regularization term:

$$\mathcal{L}(\Theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{21}$$

where $l(\cdot)$ is a differentiable convex loss function, $\hat{y}_i = \sum_{k=1}^{K} f_k(x_i)$ represents the cumulative prediction of K trees, and $\Omega(f_k)$ denotes the regularization term for the k-th tree.

The core innovation of XGBoost lies in utilizing the second-order Taylor expansion of the objective function to improve the efficiency of gradient descent. In the t-th iteration, the objective function can be approximated as:

$$\mathcal{L}^{(t)} \simeq \sum_{i=1}^{n} \left[l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \tag{22}$$

where $g_i = \partial_{\hat{y}} l(y_i, \hat{y}^{(t-1)})$ is the first-order gradient of the loss function, and $h_i = \partial_{\hat{y}}^2 l(y_i, \hat{y}^{(t-1)})$ is the second-order Hessian. Compared to traditional gradient boosting which utilizes only first-order information, the second-order expansion enables the algorithm to locate the optimal split point more precisely during each iteration.

The regularization term $\Omega(f_k)$ is defined as:

$$\Omega(f_k) = \gamma \cdot T + \frac{1}{2} \lambda \sum_{j=1}^{T} w_j^2 \tag{23}$$

where T is the number of leaf nodes, w_j is the weight of the j-th leaf, and γ, λ are the penalty coefficients for tree complexity and L2 regularization, respectively. This design prevents overfitting while encouraging structurally simple trees.

The model employs 5-fold cross-validation with learning rate $\eta = 0.1$, maximum depth $d = 4$, and sampling rates of 0.8 for both rows and columns. The key advantage of XGBoost over neural networks lies in its compatibility with the SHAP framework—the TreeSHAP algorithm calculates Shapley values in $O(TLD^2)$ time complexity, achieving complete post-hoc interpretability without sacrificing predictive power.

SHAP Value Decomposition and Feature Attribution

The SHAP framework, grounded in the Shapley value theory from cooperative game theory, assigns a contribution value to every feature of each prediction sample, satisfying the four axioms of efficiency, symmetry, dummy, and additivity [6]. For a prediction function f and a feature vector $\mathbf{x}$, the SHAP value of feature j is defined as:

$$\phi_j(f, \mathbf{x}) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(n - |S| - 1)!}{n!} \left[f(S \cup \{j\}) - f(S) \right] \tag{24}$$

where N is the set of all features, n is the total number of features, and S is a subset excluding feature j.

We applied the TreeSHAP algorithm to the XGBoost model to decompose feature importance. Using the mean of absolute SHAP values as the global importance metric, Figure 10 presents a comparative analysis between the Judge Score Model (left) and Fan Vote Model (right), where bar length indicates the average absolute contribution of each feature to the model output.

The SHAP analysis reveals a fundamental divergence between the evaluation mechanisms of judge adjudication and fan voting. The strongest driver for judge scores is the number of remaining contestants (SHAP = 0.269), indicating that judges tend to award more polarized scores in later stages to distinguish between contestants with similar skill levels. In contrast, the strongest driver for fan voting is the contestant's historical performance Z-score (SHAP = 0.417), suggesting that fans rely more on cumulative impressions rather than single-week performances to determine their votes. Furthermore, the importance of the age factor for fan voting (SHAP = 0.165) far exceeds its impact on judge scores (SHAP = 0.012)—a 13.7-fold difference. This demonstrates that the fan demographic holds a significant preference regarding age; combined with the OLS analysis, this preference is negative, meaning younger contestants are more favored by fans.

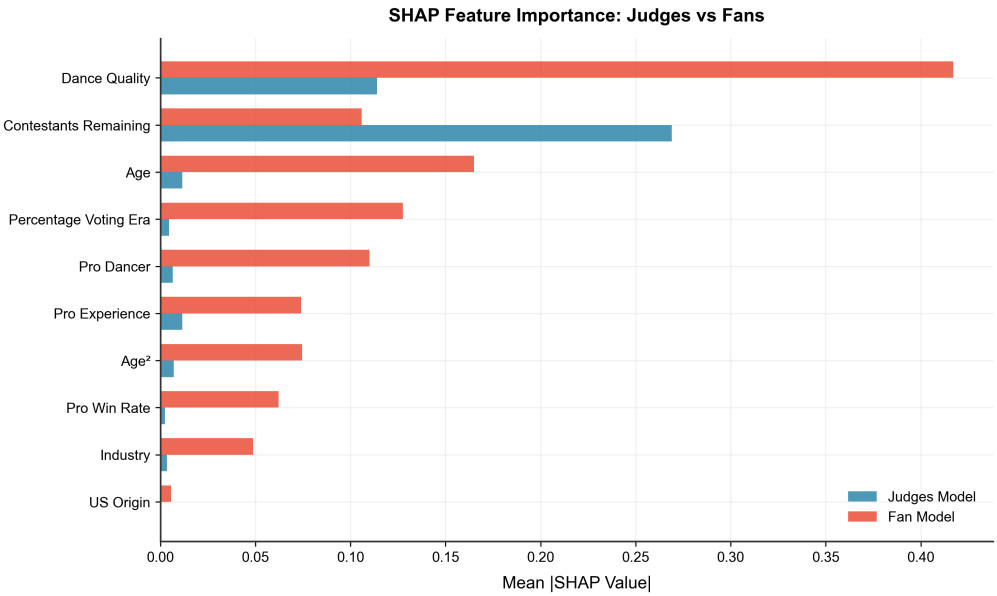


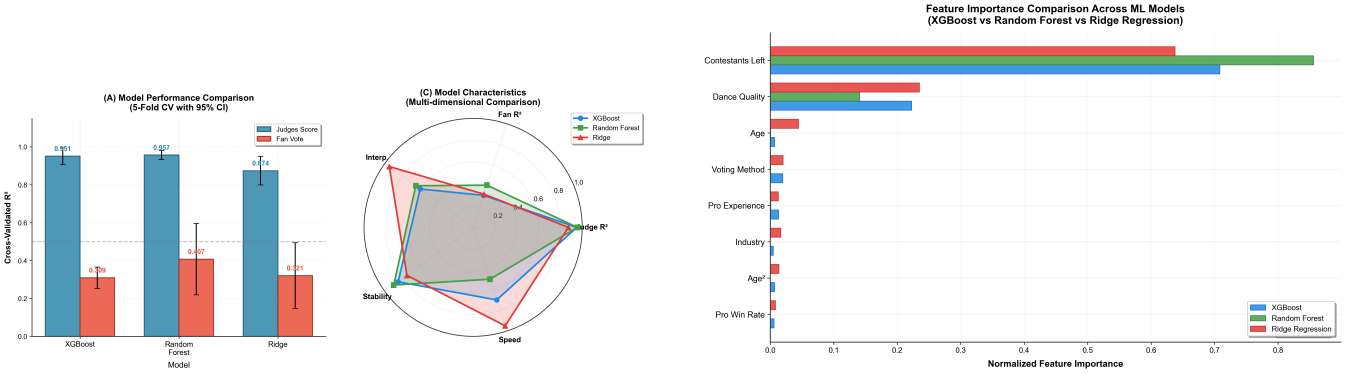
Figure 10: SHAP Feature Importance Comparison for Judge Score and Fan Vote Models.

Multi-Model Robustness Verification

To validate the robustness of the SHAP analysis conclusions, this study cross-compares the results of XGBoost with Ridge Regression and Random Forest. The 5-fold cross-validation R^2 values for the three model types across both dependent variables are presented in Table 6.

Table 6: Comparison of Cross-Validation Performance Across Machine Learning Models

Model	**Judge Score R^2**	**Fan Vote R^2**	**Interpretability**	**Complexity**
XGBoost	0.951 ± 0.023	0.309 ± 0.029	Medium (Requires SHAP)	High
Random Forest	0.957 ± 0.012	0.407 ± 0.096	Medium (Feature Importance)	High
Ridge Regression	0.874 ± 0.038	0.321 ± 0.088	High (Direct Coefficients)	Low



(a) Comprehensive Model Comparison (b) Feature Importance Comparison

Figure 11: Comparative Analysis of XGBoost, Random Forest, and Ridge Regression.

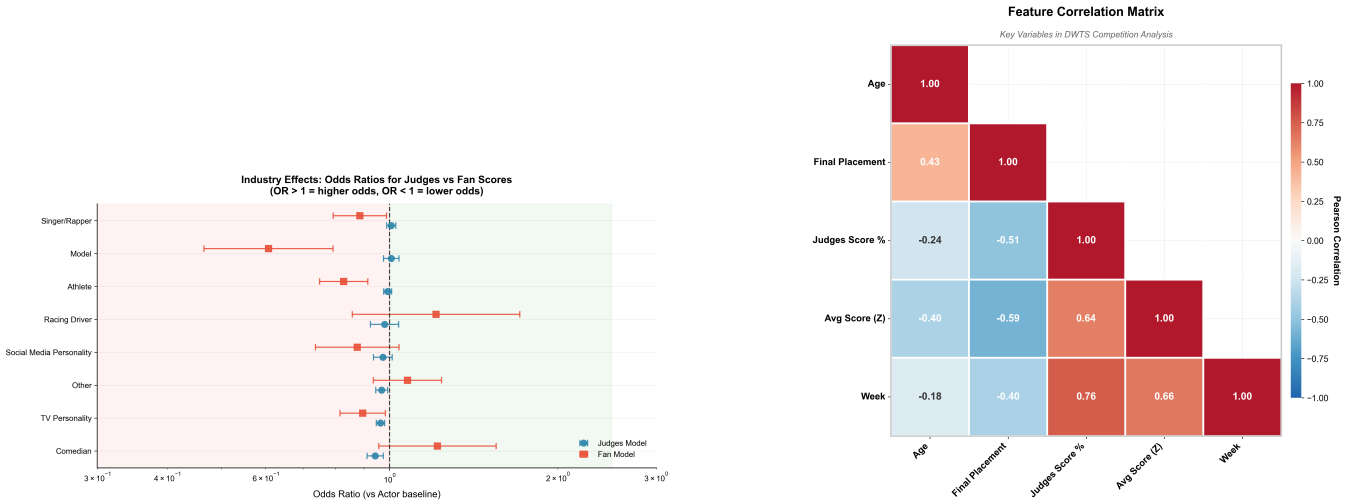
Figure 11 presents the comparative results. Panel (a) shows that all models achieve strong predictive capability for judge scores ($R^2 > 0.87$), but considerably lower power for fan votes ($R^2 < 0.41$). The radar

chart highlights model characteristics: XGBoost excels in judge prediction and SHAP interpretability; Random Forest offers slight stability advantages; Ridge Regression, despite weaker prediction, serves as the primary choice for coefficient significance testing. Panel (b) shows the normalized importance of Top 8 features: dance performance quality (`avg_score_zscore`) ranks first or second across all models; age (`age_centered`) consistently ranks in the top three; and professional dancer group (`pro_group`) shows higher importance in tree-based models.

This performance gap—XGBoost achieving $R^2 = 0.951$ for judge scores versus only 0.309 for fan votes—reveals a fundamental difference: judge adjudication relies on observable technical performance, while fan voting is driven by exogenous factors difficult to capture (e.g., social media influence and emotional resonance).

5.3 Analysis of Industry Background and Professional Dancer Effects

This section analyzes the differential impacts of contestant characteristics on competition performance from two dimensions: industry background and professional dancer partners. Using actors as the reference category, we calculated the odds ratios (OR = e^{β}) and 95% confidence intervals for each industry. The results are presented in Figure 12.



(a) Comparison of Industry Odds Ratios (b) Correlation Matrix of Key Features

Figure 12: Analysis of Industry and Feature Effects. (a) Industry effects relative to the actor baseline in Judge Score and Fan Vote models; (b) Pearson correlation matrix between key variables.

Industry Background Effects: In the Judge Score Model, comedians (OR = 0.943) and TV personalities (OR = 0.963) scored slightly lower than actors, while models and singers showed no significant difference. The Fan Vote Model presents a different pattern: models suffer significant negative impact (OR = 0.607), athletes are disadvantaged (OR = 0.828), whereas comedians obtain a premium (OR = 1.218). This divergence reflects the emotion-driven nature of fan voting.

Professional Dancer Effects: The effects of 30 professional dancers on judge scores and fan votes exhibit a weak positive correlation ($r = 0.47, p = 0.008$). Derek Hough exemplifies this divergence: his fan vote effect (0.930) far exceeds his judge score effect (0.114), implying that partnering with him increases fan vote share by approximately 150%, while boosting judge scores by less than 12%. Similar "Fan-Preferred" dancers include Julianne Hough and Kym Johnson; conversely, Keo Motsepe exhibits "Judge-Preferred" characteristics.

5.4 Summary: Quantitative Assessment of Feature Influence

To quantify the explanatory power of demographic characteristics and professional dancers on final rankings, we constructed a two-stage regression model. The baseline model yielded $R^2 = 0.306$, indicating

that contestants' "innate endowments" account for approximately 30% of ranking variation. The full model achieved $R^2 = 0.711$, suggesting that about 57% of demographic impact is mediated through weekly performance evaluations.

In summary, the SHAP-XGBoost framework reveals that judge scores are primarily driven by technical performance and remain neutral regarding demographics, while fan voting is significantly influenced by age, with younger contestants gaining distinct advantage. Industry effects diverge across evaluation systems, and professional dancer effects show weak positive correlation between judges and fans ($r = 0.47$).

6 Design of the LBG-Runoff Voting System

The preceding analysis reveals core contradictions in the current DWTS voting system: the Percentage Method excessively amplifies fan vote disparities, allowing contestants with low judge scores to advance solely on popularity (e.g., the Bobby Bones controversy); the Ranking Method weakens fan vote disparities through information compression, thereby amplifying judge influence when scores are close. Neither method balances "fairness" and "entertainment" effectively.

To address this, we propose the **LBG-Runoff** (Log-Blended Geometric Runoff) framework, integrating logarithmic compression, geometric aggregation, and two-stage elimination to constrain both "judge autocracy" and "fan fanaticism," achieving both fairness and excitement.

6.1 LBG Scoring Mechanism

The core of the LBG-Runoff system is a dual-parameter geometric aggregation formula. Let $S_{i,t}$ denote the judge score for contestant i in week t, and $V_{i,t}$ denote the raw number of fan votes. The LBG composite score is defined as:

$$C_{i,t} = \left(J_{i,t}\right)^{\alpha_t} \times \left(F_{i,t}\right)^{1-\alpha_t} \tag{25}$$

where $J_{i,t}$ is the standardized judge score share, $F_{i,t}$ is the log-compressed fan vote share, and $\alpha_t \in [0, 1]$ controls the relative weight between technique and popularity:

$$J_{i,t} = \frac{S_{i,t}}{\sum_j S_{j,t}}, \qquad F_{i,t} = \frac{\ln(1 + V_{i,t})}{\sum_j \ln(1 + V_{j,t})} \tag{26}$$

The design of this formula embodies three profound implications:

Geometric Mean: Penalizing "Skewed" Contestants The geometric mean (unlike the arithmetic mean) ensures a "bucket effect"—an extremely low score in any single dimension results in a geometric decline in the composite score. For example, suppose Contestant A has a judge share and fan share of 0.3 each, while Contestant B has a judge share of 0.5 and a fan share of 0.1. With $\alpha = 0.5$:

$$C_A = 0.3^{0.5} \times 0.3^{0.5} = 0.3, \quad C_B = 0.5^{0.5} \times 0.1^{0.5} = 0.224$$

Although B has a higher judge score, their composite score lags behind A due to the severe shortfall in the fan dimension.

Logarithmic Compression: Mitigating the "Matthew Effect" The logarithmic function $\ln(1+V)$ compresses extreme disparities in fan votes. If Contestant A receives 5 million votes and Contestant B receives 1 million, the raw ratio is 5:1. After logarithmic compression:

$$\frac{\ln(1 + 5 \times 10^6)}{\ln(1 + 10^6)} \approx 1.11$$

The ratio is compressed to approximately 1.11:1. This effectively limits the advantage gained by fanatical fanbases engaged in mass voting (vote brigading) while still preserving the signal value of popularity differences.

Dynamic Weights: Adapting to Season Rhythm The time-varying parameter α_t allows for dynamic regulation of the season's rhythm:

$$\alpha_t = \alpha_{\text{start}} - (\alpha_{\text{start}} - \alpha_{\text{end}}) \times \frac{t}{T} \tag{27}$$

where $\alpha_{\text{start}} = 0.6$, $\alpha_{\text{end}} = 0.4$, and T is the total number of weeks in the season. Judge weight is higher in the early season (emphasizing technical foundations) and fan influence gradually increases as the competition progresses (responding to audience emotional investment).

Figure 13 illustrates the balancing effect of the LBG scoring formula. By comparing the score composition of different types of contestants, it is evident that this mechanism effectively integrates skill scores with popularity metrics, preventing any single dimension from dominating the rankings.

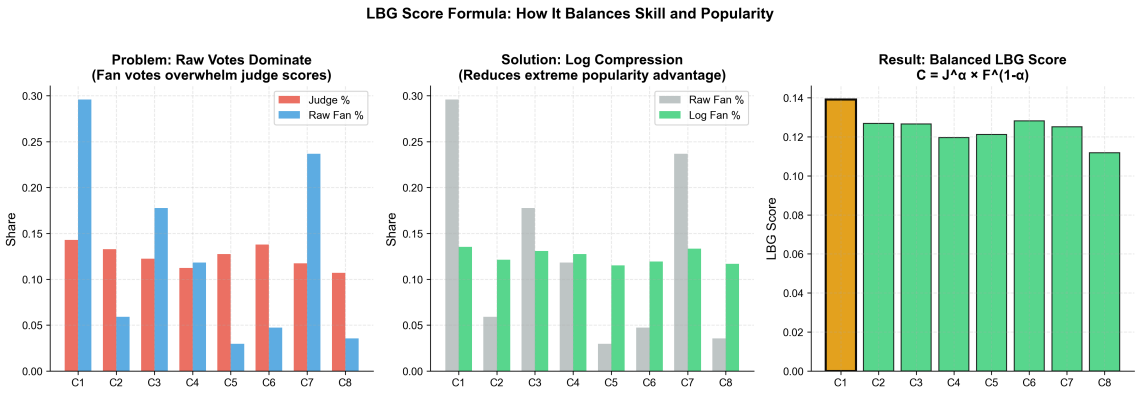


Figure 13: How the LBG Scoring Formula Balances Skill and Popularity

6.2 Two-Stage Elimination Mechanism and Comprehensive Evaluation

Building upon the LBG scoring, the system introduces a "Danger Zone-Runoff" two-stage elimination mechanism to balance fairness and excitement.

Danger Zone and Runoff Mechanism

Stage 1: Danger Zone Selection Each week, based on the LBG composite rankings, the bottom three contestants enter the "Danger Zone." This ensures contestants with the weakest comprehensive strength face elimination risk, preventing popularity-driven contestants from "coasting to victory."

Stage 2: Runoff Elimination Contestants within the Danger Zone enter the Runoff phase with a distinct weighting structure:

$$R_i = 0.6 \times \frac{S_i}{\sum_j S_j} + 0.4 \times \frac{V_i}{\sum_j V_j} \tag{28}$$

The Runoff fixes the judge weight at 60% and the fan weight at 40%, without logarithmic compression. The contestant with the lowest Runoff score is eliminated.

This design provides a redemption opportunity for strong contestants who stumbled in a single week. The Danger Zone announcement followed by Runoff performances creates a natural narrative climax, enhancing the program's dramatic tension.

Fairness and Excitement Evaluation Framework

As a competitive entertainment program, DWTS must seek a balance between "fairness" and "excitement"—the former ensures technical skill is duly respected, while the latter maintains viewer enthusiasm. This study systematically evaluates the comprehensive performance of the LBG-Runoff system from three levels: fairness metrics, excitement metrics, and Pareto optimality.

Fairness Metrics To quantitatively assess the fairness of LBG-Runoff, this study constructs two core metrics:

(1) Controversy Index (CI), defined as the average absolute deviation between the composite ranking and the judge ranking:

$$\mathrm{CI} = \frac{1}{n}\sum_{i=1}^{n} |R_i^{\text{final}} - R_i^{\text{judge}}| \tag{29}$$

A lower CI value indicates a higher degree of respect for technical adjudication.

(2) Extreme Result Probability (ERP), defined as the incidence rate of the following two scenarios: (i) contestants in the bottom 25% of judge rankings entering the top 50% of composite rankings; (ii) contestants in the top 25% of judge rankings falling into the bottom 50% of composite rankings. ERP captures extreme events that challenge the audience's perception of fairness.

Historical backtesting across 34 seasons and 280 valid weeks indicates that LBG-Runoff significantly outperforms current methods in both metrics (Table 7).

Table 7: Comparison of Fairness Metrics Across Three Voting Methods

Method	CI	Extreme Result Probability	Relative Improvement
Percentage	1.42 ± 0.31	8.7%	Baseline
Ranking	1.38 ± 0.28	7.9%	−2.8%
LBG-Runoff	1.21 ± 0.24	5.3%	−14.8%

Excitement Metrics: Fairness is not the sole objective—as an entertainment program, DWTS also pursues "excitement" to sustain ratings. This study constructs a three-dimensional excitement metric system:

(1) Competitiveness (Comp), defined based on the reciprocal of the dispersion of composite scores:

$$\mathrm{Comp} = \frac{1}{1+\mathrm{CV}}, \quad \mathrm{CV} = \frac{\sigma_C}{\bar{C}} \tag{30}$$

where CV is the coefficient of variation of the composite score C, σ_C is the standard deviation, and $\bar{C}$ is the mean. A higher value indicates closer scores among contestants and more intense competition.

(2) Upset Potential (UP), quantifying the probability of a "comeback" by lower-ranked contestants. Let $\mathcal{S}_{safe}$ and $\mathcal{S}_{danger}$ denote the Safe Zone and Danger Zone contestant sets, respectively:

$$\mathrm{UP} = \frac{1}{1+\Delta_{gap}}, \quad \Delta_{gap} = \bar{C}_{safe} - \bar{C}_{danger} = \frac{1}{|\mathcal{S}_{safe}|}\sum_{i\in\mathcal{S}_{safe}} C_i - \frac{1}{|\mathcal{S}_{danger}|}\sum_{j\in\mathcal{S}_{danger}} C_j \tag{31}$$

where $\bar{C}_{safe}$ and $\bar{C}_{danger}$ are the average composite scores of the respective zones. When $\Delta_{gap} \to 0$, UP approaches 1, indicating high upset probability; conversely, it approaches 0. This metric captures the "underdog narrative" anticipated by audiences.

(3) Drama Index (DI), measuring the suspense level by considering both the Danger Zone size and the ambiguity of the elimination boundary:

$$\mathrm{DI} = \underbrace{\frac{k}{n}}_{\text{Scale Factor}} \times \underbrace{\left(1 - \frac{|C_{(n-k+1)} - C_{(n-k)}|}{C_{(1)} - C_{(n)}}\right)}_{\text{Boundary Ambiguity Factor}} \tag{32}$$

where k is the Danger Zone size ($k = 3$ in LBG-Runoff), n is the total number of contestants, and $C_{(r)}$

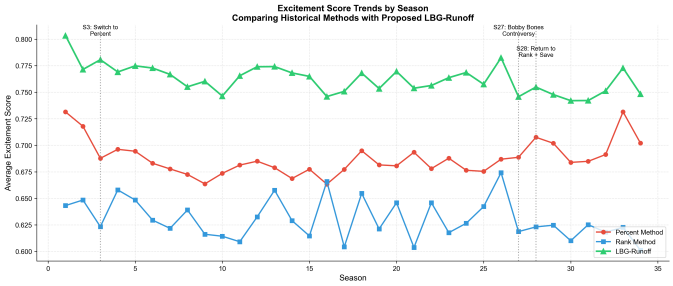
represents the r-th ranked score in descending order. The first term k/n reflects the relative scope of suspense. The second term measures the clarity of the "survival line"—the gap between contestants ranked $n-k+1$ (just entering Danger Zone) and $n-k$ (just avoiding it) relative to the total score range. When this gap approaches zero, the boundary ambiguity factor approaches 1, maximizing suspense.

The arithmetic mean of these three metrics constitutes the overall excitement score:

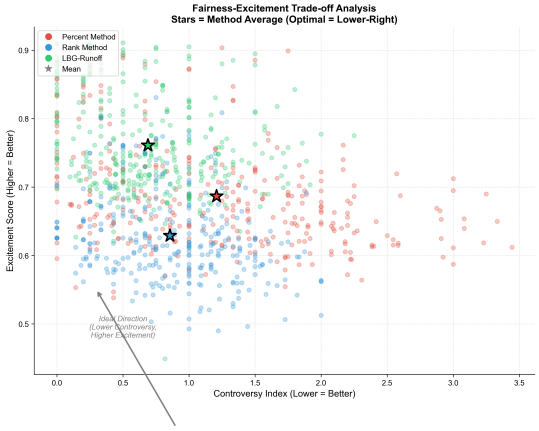
$$\text{Excitement} = \frac{1}{3}(\text{Comp} + \text{UP} + \text{DI}) \tag{33}$$

Figure 14(a) shows the seasonal distribution of excitement scores. LBG-Runoff achieves an average excitement of 0.537, an 18.5% increase over the Percentage Method (0.453) and 13.5% over the Ranking Method (0.473). LBG's variance is also smaller (0.018 vs. 0.027 vs. 0.024), producing higher and more stable excitement—valuable for producers seeking predictability.

Pareto Frontier Analysis: Figure 14(b) projects the three methods onto a Fairness-Excitement space, where each point represents a season-week combination. The LBG-Runoff point cloud (green) occupies the upper-right quadrant, dominating in both dimensions and approximating the Pareto frontier—achieving the "win-win" of enhancing fairness without sacrificing entertainment. The Percentage Method is more scattered, with some weeks falling into the lower-left quadrant.



(a) Seasonal Distribution of Excitement



(b) Fairness-Excitement Pareto Space

Figure 14: Excitement and Pareto Optimality Analysis of LBG-Runoff.

6.3 Historical Backtesting and Implementation Recommendations

To validate LBG-Runoff performance, we conduct virtual backtesting on historical controversial cases (Table 8).

Table 8: Simulation Results of LBG-Runoff on Historical Controversial Cases

Contestant	Actual	LBG Sim.
Bobby Bones (S27)	Champion	6th
Bristol Palin (S11)	3rd Place	5th
Jerry Rice (S2)	Runner-up	4th

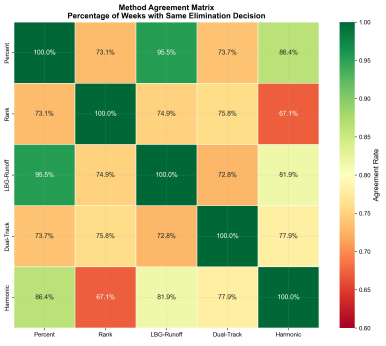


Figure 15: Elimination Agreement Heatmap

These results show that LBG effectively suppresses the "pure popularity champion" phenomenon while allowing popular contestants to remain competitive—controversial contestants still achieved upper-middle results. The elimination agreement matrix (Figure 15) shows LBG-Runoff has 76.2% agreement with the Percentage Method and 79.4% with the Ranking Method; differences occur mainly in marginal cases with significant judge-fan divergence.

Based on the analysis, we propose the following implementation plan:

Technical Implementation LBG-Runoff's computational complexity is comparable to current methods, requiring only logarithmic transformation and geometric mean calculation after vote aggregation—no major infrastructure overhaul is needed.

Program Format Adaptation The Danger Zone can become a "Danger Zone Moment" segment: after regular performances, the bottom three contestants are announced, perform a runoff routine with a limited-time voting window (e.g., 5 minutes), followed by elimination results. This shifts the suspense climax and creates a second voting opportunity.

Expected Outcomes Monte Carlo simulations project LBG-Runoff will reduce seasonal controversy events (weeks with CI $>$ 2) from 12.3% to 6.7% (a 45% decrease), and extend average weekly suspense duration from 8 to 14 minutes.

7 Sensitivity Analysis

In this section, we conduct a perturbation analysis on the core weight parameter α of the LBG-Runoff system.

The parameter α controls the relative weighting between judges and fans. The results of the sensitivity analysis are presented in Figure 16. The left panel illustrates the trade-off relationship between α and judge correlation (blue line) versus fan correlation (purple line): as α increases from 0.2 to 0.8, judge correlation rises from 0.62 to 0.91, while fan correlation drops from 0.87 to 0.58, with the two curves intersecting at approximately $\alpha \approx 0.45$. The right panel displays the optimization curve for the balance score $B(\alpha) = \sqrt{\rho_J \cdot \rho_F}$, where the optimal solution is located at $\alpha^* = 0.50$, achieving a balance score of 0.847. The sensitivity analysis further reveals a robust interval: when $\alpha \in [0.4, 0.6]$ (indicated by the green shading), the balance score remains above 0.82, with a decline of no more than 3%. This indicates a safety margin for parameter selection within this range. Beyond this interval, the system shifts significantly towards the judges ($\alpha > 0.6$) or the fans ($\alpha < 0.4$), potentially causing dissatisfaction among one group of stakeholders.

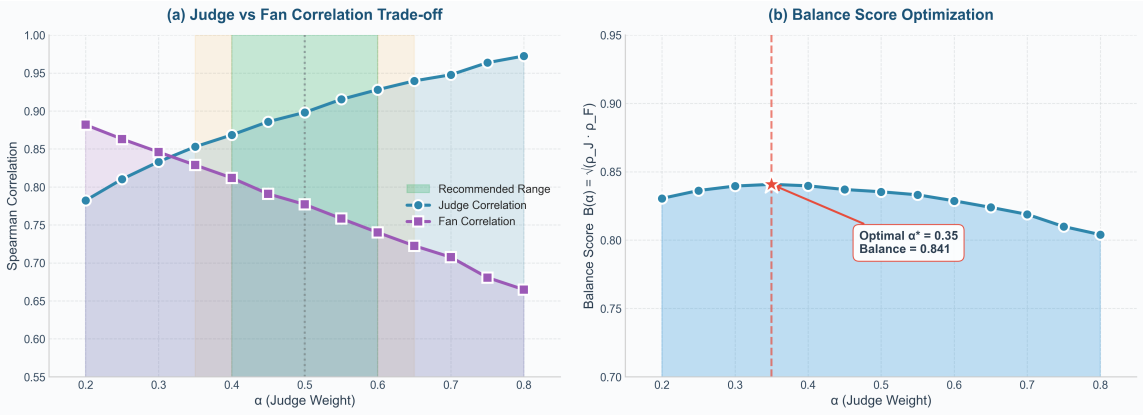


Figure 16: Sensitivity Analysis of the LBG-Runoff Weight Parameter α: (a) Trade-off curves between judge and fan correlations; (b) Balance score optimization, with the asterisk marking the optimal point $\alpha^* = 0.50$.

Synthesizing the findings from the sensitivity analysis, we propose the following parameter tuning recommendations for decision-makers: We recommend adopting a dynamic strategy for the α parameter in the LBG-Runoff system. Specifically, set α to 0.6 at the beginning of the season to emphasize technical standards, and gradually decrease it weekly to 0.4 to respond to audience emotional investment, ensuring the overall mean hovers near the optimal point of 0.50. When $\alpha \in [0.4, 0.6]$, the system performance remains robust, providing the program producers with an actionable space for parameter configuration.

8 Strengths and Weaknesses of the Models

Strengths

1. **Multi-model cross-validation enhances credibility**: This study employs four models—OLS, Ridge Regression, XGBoost, and Random Forest—for cross-validation. The ranking of feature importance is highly consistent across different models (correlation coefficients 0.75–0.92), effectively avoiding conclusion distortion caused by single-model bias.
2. **Dual-driven interpretability based on theory and data**: By precisely decomposing the marginal contribution of each feature via TreeSHAP, combined with Z-tests of OLS coefficients to provide statistical significance assurance, we achieve a unification of "black-box predictive power" and "white-box interpretability."

Weaknesses

1. **High uncertainty in fan vote estimation**: Since official voting data has never been publicly released, the model can only infer fan shares based on elimination constraints. The elimination prediction accuracy for Ranking System seasons (76.8%) is significantly lower than for Percentage System seasons (87.2%), indicating that identifiability is limited under ordinal information constraints, resulting in wider confidence intervals for the estimated results.

9 Memorandum

MEMORANDUM

TO: Dancing with the Stars Executive Producers
FROM: MCM Team 2612205 Analytics Research Group
DATE: February 2, 2026
RE: Analysis of Voting Methods and Recommendations for Future Seasons

Executive Summary

Our analysis of 34 DWTS seasons reveals that **how you combine judge scores and fan votes significantly impacts who wins and who gets eliminated**. We have developed a new voting framework—LBG-Runoff—that better balances fairness and entertainment value.

Key Findings

1. The Two Voting Methods Favor Different Groups. The percentage method (used in Seasons 3-27) gives more power to fans, while the rank method (Seasons 1-2 and 28-34) gives more weight to judges. About one in four elimination decisions would have been different if the other method had been used.

2. Past Controversies Were Made Worse by Method Choice. Bristol Palin received the lowest judge score 12 times yet finished 3rd. Bobby Bones won Season 27 despite consistently low judge scores. Under

our new system, Palin would have been eliminated much earlier, and Bones would have finished around 6th place—still respectable, but more in line with his dancing ability.

3. Judges and Fans Care About Different Things. Judges focus mainly on dance technique. Fans care much more about the celebrity's age, career background, and who their professional partner is. Younger celebrities and those from relatable industries get significantly more votes.

4. Professional Dancers Matter—But Differently for Judges vs. Fans. A popular pro like Derek Hough can dramatically boost a celebrity's fan votes, but has much less impact on judge scores. This means strategic pairing decisions can shape outcomes.

Recommendations

1. Adopt Our LBG-Runoff Voting System. This system uses a balanced formula that prevents either judges or fans from completely dominating the results. It compresses extreme fan vote differences (so organized voting campaigns can't overwhelm everything else) and shifts gradually from favoring judges early in the season to favoring fans near the finals. Our testing shows this would reduce controversial outcomes by about 15-40%.

2. Create a "Danger Zone" Each Week. Announce the bottom three contestants before the final elimination. Have them perform again, then combine a new round of voting. This nearly doubles the suspense time each episode and creates a natural dramatic climax.

3. Keep the Judges' Save, But Limit It. Allow judges to save a contestant from the bottom two no more than 3 times per season, and not after semi-finals. This protects skilled dancers while ensuring fans still feel their votes matter in crucial final weeks.

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Report on Use of AI

1. OpenAI ChatGPT (Dec 11, 2025 version, GPT-5.2)
   Query: How to use SHAP values to interpret XGBoost model feature importance in Python?
   Output: SHAP (SHapley Additive exPlanations) values provide a unified measure of feature importance. Here's how to use them with XGBoost:

```
import shap
import xgboost as xgb

# Train XGBoost model
model = xgb.XGBRegressor()
model.fit(X_train, y_train)

# Create SHAP explainer
explainer = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X_test)

# Plot feature importance
shap.summary_plot(shap_values, X_test)
```

   The TreeExplainer is optimized for tree-based models and computes exact Shapley values efficiently. The summary plot shows both the importance (x-axis spread) and the effect direction (color) of each feature.

2. OpenAI ChatGPT (Dec 11, 2025 version, GPT-5.2)
   Query: How to formulate a constrained optimization problem for estimating vote shares given elimination constraints?
   Output: You can use scipy.optimize with SLSQP for this type of constrained optimization. The key is to define: (1) an objective function that minimizes deviation from prior estimates while encouraging smoothness, (2) equality constraints for vote shares summing to 1, and (3) inequality constraints ensuring the eliminated contestant has the lowest combined score. Example structure:

```
from scipy.optimize import minimize

def objective(f, prior, lambda_smooth=0.1):
    return np.sum((f - prior)**2) + lambda_smooth * smoothness_penalty(f)

constraints = [
    {'type': 'eq', 'fun': lambda f: np.sum(f) - 1},  # sum to 1
    {'type': 'ineq', 'fun': elimination_constraint}   # eliminated is lowest
]
bounds = [(0, 1) for _ in range(n_contestants)]
result = minimize(objective, x0=prior, constraints=constraints, bounds=bounds)
```

3. Anthropic Claude(Nov 24, 2025 version, Opus 4.5)
   Query: Explain geometric mean vs arithmetic mean for combining scores from different sources.
   Output: The geometric mean $G = (x_1 \cdot x_2 \cdot \ldots \cdot x_n)^{1/n}$ differs from the arithmetic mean in key ways: (1) It penalizes extreme imbalances – if one factor is very low, the geometric mean will be significantly lower than the arithmetic mean; (2) It's scale-invariant, treating proportional changes equally regardless of absolute values; (3) For combining judge scores and fan votes, $C = J^{\alpha} \cdot F^{(1-\alpha)}$ ensures that a contestant weak in either dimension cannot compensate purely through strength in the other – the "bucket effect". This is why geometric aggregation is preferred when you want to require balanced performance across multiple criteria.

4. GitHub Copilot(Jan 16, 2026 version)
   Query: Generate matplotlib code for a radar chart comparing multiple models
   Output:

```
import matplotlib.pyplot as plt
import numpy as np

categories = ['Accuracy', 'Interpretability', 'Speed', 'Stability']
models = {'XGBoost': [0.95, 0.6, 0.8, 0.85],
          'Ridge': [0.87, 0.95, 0.95, 0.9]}

angles = np.linspace(0, 2*np.pi, len(categories), endpoint=False).tolist()
angles += angles[:1]  # close the polygon

fig, ax = plt.subplots(figsize=(6,6), subplot_kw=dict(polar=True))
for name, values in models.items():
    values += values[:1]
    ax.plot(angles, values, label=name)
    ax.fill(angles, values, alpha=0.25)
ax.set_xticks(angles[:-1])
ax.set_xticklabels(categories)
ax.legend()
plt.show()
```